

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas

Aug 2024 – May 2026

Master of Science in Computer Engineering

Fayetteville, AR

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College

Aug 2019 – May 2023

Bachelor of Engineering in Electronics and Communication

India

TECHNICAL SKILLS

Data Engineering: Flink SQL, ETL/ELT, Azure SQL, data validation, Apache Kafka, Apache Flink, Apache Airflow, ClickHouse, Databricks, Apache Spark, SQL Server, Delta Lake, PostgreSQL, MySQL

Machine Learning and AI: object detection and tracking, RAG, PyTorch, Scikit-learn, computer vision, LangChain, Pandas, NumPy, LSTM, FAISS, Ollama, Llama 3.2, Streamlit, Flask

Cloud and Tools: AWS Glue, AWS Lambda, LabVIEW, NI TestStand, Git, Amazon EMR, Amazon S3, Docker, Linux, Jira, Grafana

Languages: SQL, Python, C, C#, C++, Java

PROJECTS

DualFit - Two-Stage Virtual Try-On (AICV Lab) *PyTorch, Res-UNet, flow-based warping, inpainting*

- Developed a PyTorch-based two-stage Virtual Try-On model combining flow-based garment warping with Res-UNet image synthesis, with preprocessing and inpainting pipelines to improve garment alignment and visual realism
- Trained and evaluated on the VITON-HD dataset, reaching 24.9 PSNR, 0.913 SSIM, and 5.261 FID

Expense Tracker with Predictive Analytics

C++, MySQL, Python, SQL

- Developed a cross-language expense management application integrating C++ business logic with a MySQL database for storing, retrieving, categorizing, and processing historical expense data
- Optimized SQL queries generating category-based expense summaries and reports, and implemented Python analytics over historical spending to generate next-month expense predictions

PROFESSIONAL EXPERIENCE

Data Engineering Intern

Jul 2024 – Present

InsOps Inc.

Chicago, Illinois

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant

Sep 2024 – May 2026

University of Arkansas, Dept. of Computer Science and Engineering

Fayetteville, AR

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant

Sep 2024 – Jan 2026

AICV Lab, University of Arkansas

Fayetteville, AR

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer

Jun 2023 – Jun 2024

Soliton Technologies

India

- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation